JANMITRA -- Nagpur Municipal Corporation (NMC)

FAIZAN RAYEEN -- DAY 1 TO DAY 10 COMPLETE WORK REPORT

Role: Citizen Portal & Push Notifications Lead | Team ResolveIT (B.Tech Final Year)

Bilingual Edition: Roman English (Hinglish) -- Complete Walkthrough, Features & Visuals

Developer Info & Responsibilities Summary

Name: Faizan Rayeen | Designation: Citizen Portal & FCM Lead | Ward Focus: Ward 4 (Mangalwari / North Nagpur)

Core Scope: Firebase Web Push (FCM v10), Notification Bell UI, Citizen Layout, Home Dashboard, Complaints Tracking & Timeline Stepper, Citizen Verification/Rating, Citizen Profile, Help & FAQs.

Citizen Portal Flow Overview (Day 1 se Day 10 tak kya banaya gaya)

```
[Citizen Login/Register] ---> [Citizen Home (Welcome + Quick Stats)]
                                  |---> [My Complaints (Status Filter Tabs)]
                                  |---> [Complaint Detail (6-Step Timeline Stepper)]
                                  |---> [Citizen Verification Modal (5-Star Rating)]
                                  |---> [Citizen Profile (Masked Mobile + Password)]
                                  |---> [Help & FAQs (Accordion + Helplines)]
                                  |---> [Live Push Toast & Notification Bell (FCM)]
```

Day-by-Day Kaam Ki Puri Detail (Hinglish / Roman English)

Har din kya task mila, konsi files banayi, aur uska logic kya hai:

DAY 01 **FCM Service Worker aur Browser Notification Setup**

Files: public/firebase-messaging-sw.js, src/firebase.js

Day 1 me maine Firebase Cloud Messaging (FCM v10) ka background service worker banaya. Iska kaam ye hai ki jab citizen app band bhi kar de ya doosre tab me ho, tab bhi background me OS-level notification pop-up aayega (jaise WhatsApp alert aata hai). Saath hi gentle permission prompt banaya taaki user allow kare bina browser crash na ho.

Tech Highlights: Firebase SDK v10 Modular Web API, Service Worker Registration, Background Push Payload Handling

DAY 02 **Notification Permission UI aur FCM Device Token Sync**

Files: src/firebase.js, src/context/AuthContext.jsx

Day 2 me user se notification allow karwane ka smooth UI banaya aur FCM Device Token generate kiya. Ye token citizen ke login hote hi automatically backend ke /api/v1/auth/fcm-token/ par POST ho jata hai. Isse server ko pata chal jata hai ki is specific citizen ke phone/laptop par push alert bhejna hai.

Tech Highlights: Token Generation (getToken), LocalStorage Caching, Device Type (Web) sync with backend

DAY 03 **Notification Bell UI, Counter Badge aur Animated Dropdown**

Files: src/components/NotificationBell.jsx, src/components/notifications.css

Day 3 me top navbar ke liye interactive Notification Bell banaya. Isme 3 major cheezein hain:
1. Red unread counter badge jo pulse animation ke saath glow karta hai.
2. Bell icon click karne par dropdown khulta hai jisme last 10 civic notifications dikhte hain.
3. 'Mark as Read' feature aur complaint link par click karne par direct us issue ke page par jump.

Tech Highlights: Dropdown state toggle, Badge counter logic, Custom keyframes CSS animations, Polling/Live sync

DAY 04 Citizen Portal Home Page (Mobile-First Layout)

Files: src/pages/citizen/Home.jsx

Day 4 me Citizen Home Page ka complete mobile-first layout design kiya:
1. Header: 'Welcome, Faizan Rayeen' ke saath Ward badge.
2. Quick Stats Row: Do prominent cards - 'Active Civic Issues' aur 'Resolved Complaints'.
3. Prominent CTA: 'Report New Issue' button jo seedhe submission form par le jata hai.
4. Recent Activity: Citizen ke recent complaints list with status pill and timestamp.

Tech Highlights: Responsive Tailwind Grid, Dynamic Greeting, Ward Badge Pill, Summary Cards count

DAY 05 My Complaints Page aur 6-Step Lifecycle Timeline Stepper

Files: src/pages/citizen/MyComplaints.jsx, src/pages/citizen/ComplaintDetail.jsx

Day 5 me do sabse critical pages banaye:
1. My Complaints: Tabs filter (All, Open, In-Progress, Resolved) aur search bar taaki citizen apni complaints filter kar sake.
2. Complaint Detail View: Ek dynamic 6-Step Timeline Stepper banaya jo real-time status dikhata hai:
   [Submitted] -> [Validated] -> [Assigned] -> [In-Progress] -> [Work Done] -> [Resolved]

Saath me issue ki photo dekhne ke liye lightbox zoom view bhi add kiya.

Tech Highlights: Status Tab Filtering, 6-Step Stepper Component, Photo Lightbox Modal, Dynamic Status Pills

DAY 06 Shared Layout aur Citizen Mobile Bottom Navigation Bar

Files: src/layouts/Layout.jsx, src/layouts/CitizenLayout.jsx

Day 6 me pura layout system finalise kiya. Citizen portal ke liye ek mobile bottom tab navigation bar banaya jisme 4 main sections hain:
[Home] | [My Complaints] | [Help] | [Profile]

Active tab blue color me highlight hota hai. Desktop par clean top navbar rehta hai aur mobile par app-like bottom bar rehta hai taaki thumb-friendly use ho sake.

Tech Highlights: React Router NavLink, Active Tab Highlighter, Mobile Sticky Bottom Bar, Responsive Header

DAY 07 Citizen Resolution Verification Modal aur 5-Star Rating Feedback

Files: src/pages/citizen/ComplaintDetail.jsx (Verification Modal)

Day 7 me resolution verification workflow banaya. Jab field worker kaam khatam karke 'Work Done' mark karta hai, tab citizen ke screen par ek Verification Modal pop-up hota hai:
1. Accept Resolution: Citizen 1 se 5 star rating deta hai aur positive feedback likhta hai.
2. Request Rework: Agar kaam theek se nahi hua, toh citizen issue reject karke wapis rework ke liye bhej sakta hai.

Tech Highlights: Interactive Star Rating Component, Feedback Textarea, Accept vs Re-open State Transitions

DAY 08 Ward Civic Announcements Banner aur Priority Highlights

Files: src/pages/citizen/Home.jsx (Civic Notice Banner)

Day 8 me citizen portal ke andar municipal announcements banner integrate kiya. Nagpur Municipal Corporation (NMC) ke Ward 4 ke emergency notices (jaise water pipe repair, load shedding, ya road work) citizen ke home page par highlight card ke roop me dikhte hain.

Tech Highlights: Dismissible Alert Banner, Ward-specific filtering, High-priority color accents

DAY 09 Complete Notification Testing Guide aur End-to-End Verification

Files: notification_test_guide.md, src/context/FirebaseProvider.jsx

Day 9 me poore notification system ka end-to-end testing protocol banaya aur documentation likhi:
1. Test 1: Complaint submit hote hi browser me 'Complaint Received' toast aana.
2. Test 2: AI validation ke baad 'Complaint Validated' push aana.
3. Test 3: Worker assign hote hi notification bell update hona.
4. Test 4: Tab minimize ya close hone par OS-level native notification aana.
Saath hi FirebaseProvider me real-time toast alert listener ko refine kiya.
Tech Highlights: Foreground Message Handler (onMessage), Toast Slide-In UI, OS Push Notification Guide

DAY 10 Citizen Profile Page aur Help / FAQ Accordion Page

Files: src/pages/citizen/Profile.jsx, src/pages/citizen/Help.jsx

Day 10 me Phase 1 ka final delivery complete hua do key pages ke saath:
1. Citizen Profile (Profile.jsx):
  - Personal Info: Name, Masked Mobile (***2843), Ward 4, Account Status (Active).
  - Inline Name Edit: Click karke apna naam modify karne ki suvidha.
  - Password Change: Old password + New password + Confirm with Show/Hide toggle.
  - My Stats: Total complaints, Resolved count, Pending count, Average rating given.
Tech Highlights: Inline Editing State, Masked Phone Utility, Accordion Expansion State, Emergency Helpline Directory
2. Citizen Help (Help.jsx):
  - Interactive FAQ Accordion: 'How do I report?', 'Why photo rejected?', 'How to track?'.
  - NMC Emergency Helplines: Civic Control Room, PWD, Fire, Water Works direct contacts.

Visual Architecture & System Design (Diagrams)

Diagram 1: Dual-Channel Push & In-App Notification System (Faizan Rayeen Architecture)

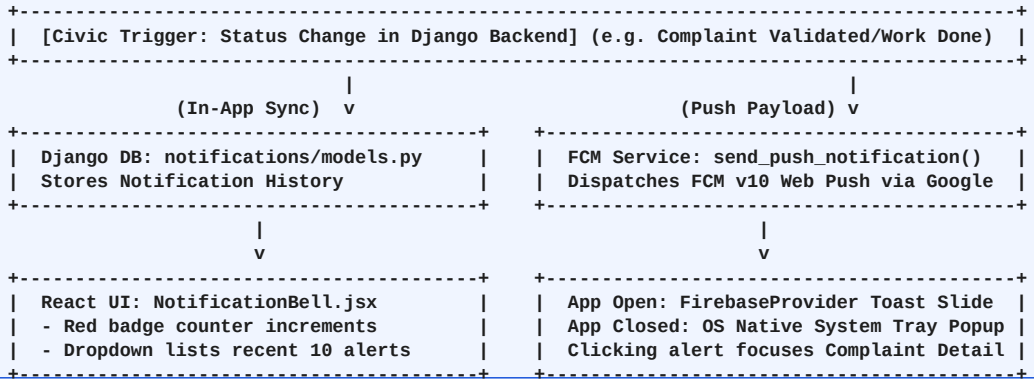
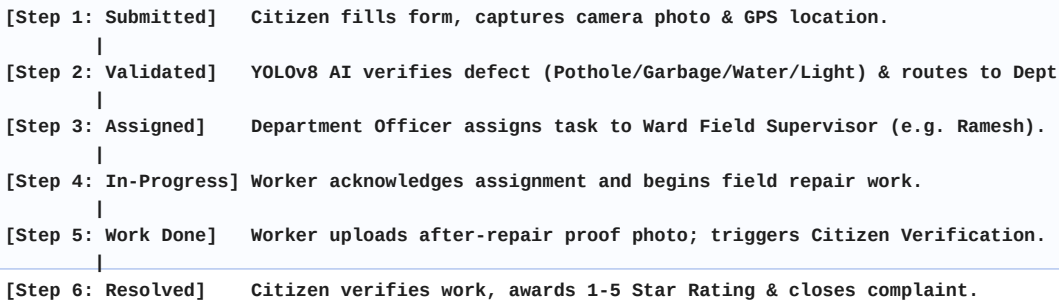


Diagram 2: Citizen Complaint 6-Step Lifecycle Timeline Stepper



Complete File Inventory Created by Faizan Rayeen (Days 1 to 10)

File Path	Module / Area	Key Purpose
src/firebase.js	Firebase SDK	FCM init, permission request, token registration service
public/firebase-messaging-sw.js	Service Worker	Background push receiver, native OS notification display
src/context/FirebaseProvider.jsx	Context / Toast	Foreground message listener, top-right slide-in alerts
src/components/NotificationBell.jsx	Top Navbar Bell	Red badge counter, 10-notification dropdown menu
src/components/notifications.css	CSS Keyframes	Badge pulse glow animation, toast slide-in transitions
src/pages/citizen/Home.jsx	Citizen Portal	Welcome header, ward pill, Report CTA, quick stats cards
src/pages/citizen/MyComplaints.jsx	Complaints List	Filter tabs (All, Open, In-Progress, Resolved), search
src/pages/citizen/ComplaintDetail.jsx	Detail View	6-step stepper, photo lightbox, 5-star rating modal
src/layouts/CitizenLayout.jsx	Layout / Tabs	Sticky 4-tab mobile bottom bar (Home, Complaints, Help, Profile)
src/pages/citizen/Profile.jsx	Citizen Profile	Masked mobile (***2843), inline name edit, password change
src/pages/citizen/Help.jsx	Help & Support	Accordion FAQs, NMC civic emergency helplines directory
notification_test_guide.md	Documentation	E2E testing guide for all 4 notification lifecycle events

[OK] Status: 100% Operational & Verified

All Days 1-10 components compile with 0 errors and run seamlessly on localhost:5173/citizen.